

A Novel Scaling Indicator of Early Warning Signals

with an application to tropical cyclones

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Early warning signals of tipping points

Tipping point: an abrupt change in a system

We want to predict the tipping point

We can use

- Variance
- Autocorrelation
- DFA
- Potential analysis
- Model fitting

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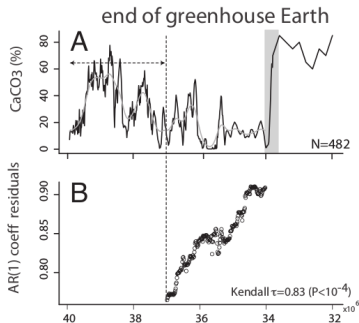
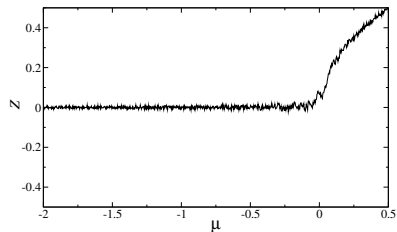
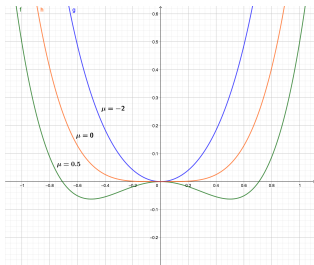


Figure: Dakos, V., Scheffer, M., van Nes, E. H., Brovkin, V., Petoukhov, V. and Held, H. (2008). Slowing down as an early warning signal for abrupt climate change. *Proc. Nat. Acad. Sci. USA* 105, 14308-14312.

The pitchfork bifurcation

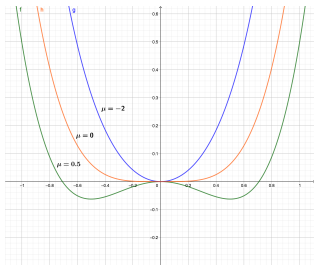
$$\dot{z} = -\frac{\partial}{\partial z} (z^4 - \mu_t z^2) + \eta_t$$



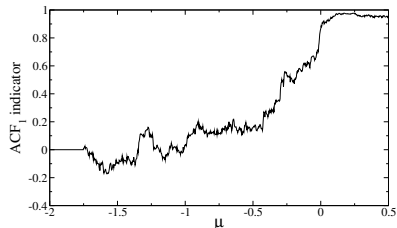
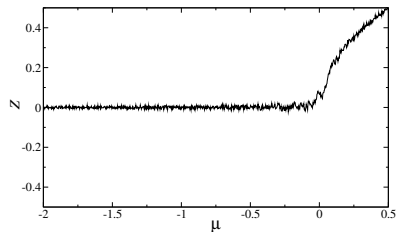
Bifurcation at $\mu = 0$.

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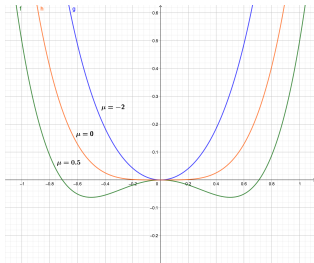


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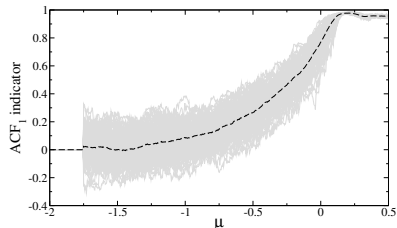
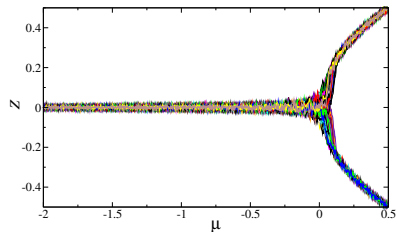


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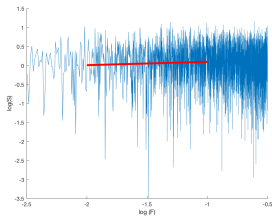
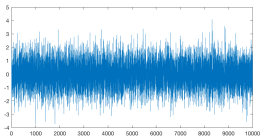


Bifurcation at $\mu = 0$.



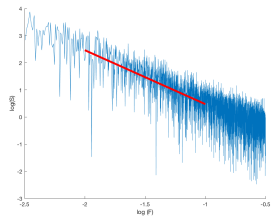
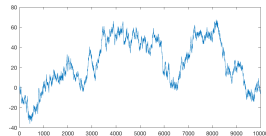
Colours of noise

White



$$\beta = -0.08$$

Red

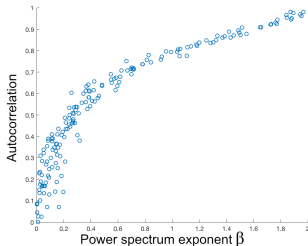


$$\beta = 1.99$$

$$S(f) \sim f^{-\beta}$$

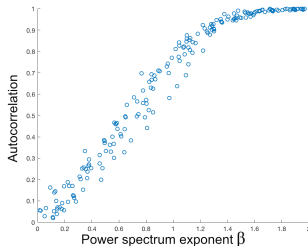
But, be careful

AR(1) model



$$x_n = \phi x_{n-1} + \eta_n$$

AR(63) model



MatLab DSP Toolbox
Kasdin, N.J. (1995)*

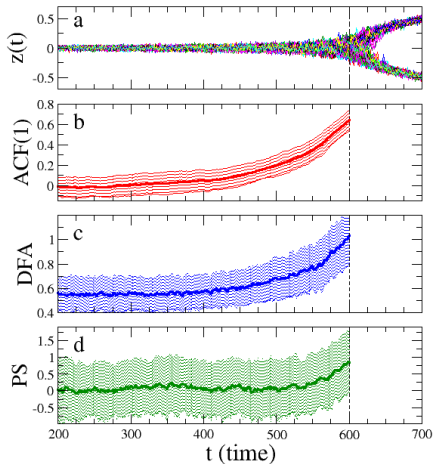
*Kasdin, N.J. "Discrete Simulation of Colored Noise and Stochastic Processes and 1/f Power Law Noise Generation." Proceedings of the IEEE, Vol. 83, No. 5, 1995, pp. 802-827.

Comparison: pitchfork bifurcation

Bifurcation occurs at $t=600$

PS-indicator definitely the worst

Indicators applied to model data

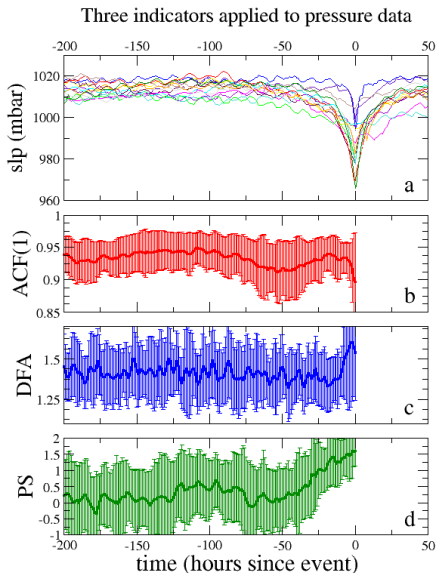


Comparison: tropical cyclones

Sea level pressure measurements taken at the landfall-location of 14 tropical cyclones.

PS-indicator works better than others

Atlantic hurricanes Andrew (1992), Opal (1995), Floyd (1999), Charley (2004), Frances (2004), Jeanne (2004), Katrina (2005), Rita (2005), Ivan (2006) and Ernesto (2006); typhoons Zeb (1998), Megi (2010) and Flo (1990); and Cyclone Hudhud (2014)



Thank you!

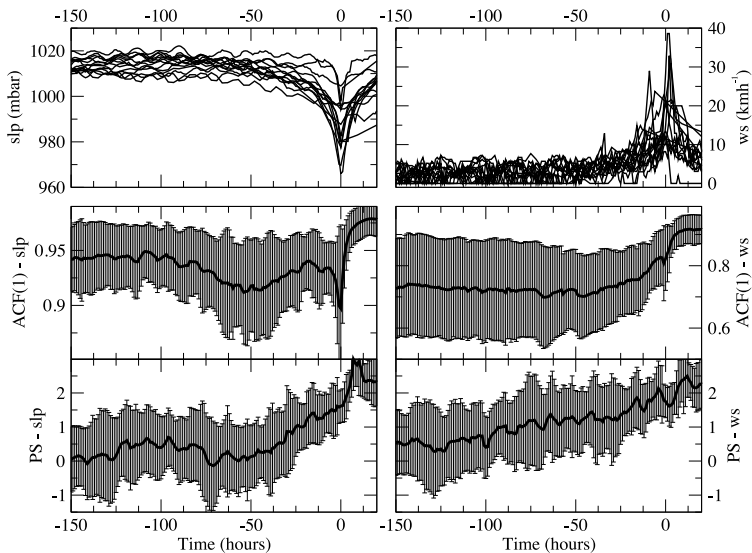
Our paper:

- Prettyman, J., Kuna, T., Livina, V. (2018). A novel scaling indicator of early warning signals helps anticipate tropical cyclones. EPL (Europhysics Letters), 121(1), 10002.

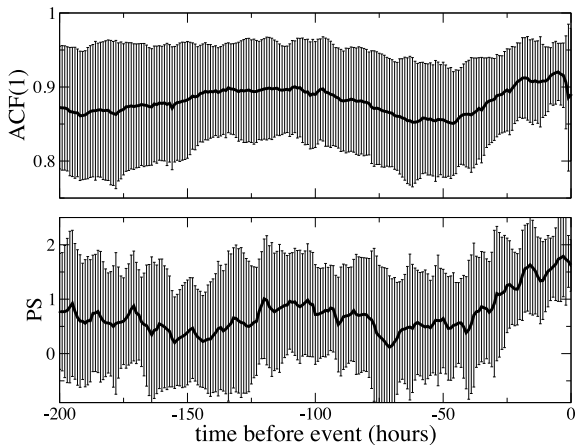
Review of methods:

- Livina, V. N., Ditlevsen, P. D., Lenton, T. M. (2012). An independent test of methods of detecting system states and bifurcations in time-series data. Physica A: Statistical Mechanics and its Applications, 391(3), 485-496.

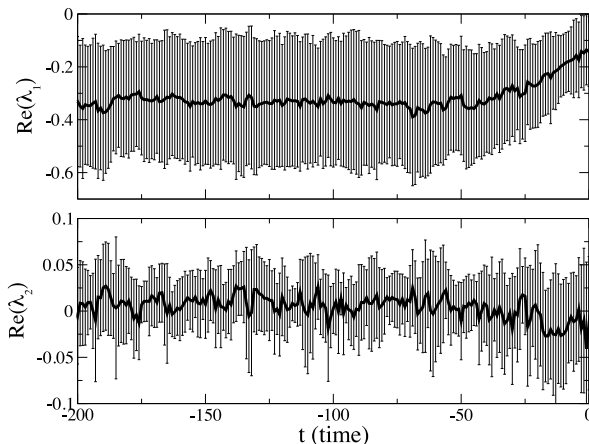
Two variables



Two variables: EOF method



Two variables: Williamson and Lenton method



Williamson, M. S., Lenton, T. M. (2015). Detection of bifurcations in noisy coupled systems from multiple time series. *Chaos: An Interdisciplinary Journal of Nonlinear Science*, 25(3).